

HASTINGS, NE, USA

WMO: 725525

Lat: 40.601N

Lon: 98.426W

Elev: 1961

StdP: 13.68

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94949

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.8	2.8	-9.3	3.6	-0.1	-5.5	4.4	4.6	34.7	27.3	30.4	28.6	10.1	340	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	95.1	73.1	92.1	72.9	89.7	72.3	77.3	89.1	75.7	87.2	74.2	85.6	11.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.5	133.9	83.7	72.3	128.7	82.6	70.5	120.8	80.7	42.2	89.3	40.5	87.3	39.0	85.3	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.4	24.9	21.2	-9.4	100.3	4.0	2.5	-12.3	102.1	-14.7	103.5	-17.0	104.9	-19.9	106.7		
			-10.0	80.4	4.0	2.0	-12.8	81.8	-15.1	83.0	-17.4	84.1	-20.2	85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	26.2	29.4	40.5	50.9	61.6	72.3	77.1	74.1	66.6	53.1	40.0	28.6		
	DBStd	20.26	12.04	12.37	11.62	9.78	8.69	6.71	5.83	5.56	8.14	9.62	10.31	10.81		
	HDD50	2836	738	579	336	104	11	0	0	0	2	76	326	664		
	HDD65	5925	1203	996	761	432	172	16	2	4	77	382	751	1129		
	CDD50	3499	0	3	41	130	369	670	839	748	499	174	26	0		
	CDD65	1110	0	0	1	8	65	236	376	286	124	14	0	0		
	CDH74	11277	0	0	28	148	719	2403	3959	2596	1239	177	8	0		
	CDH80	4667	0	0	4	37	240	980	1857	1041	459	48	1	0		
	WSAvg	10.1	10.3	10.7	11.6	12.6	11.0	9.8	8.2	7.9	9.1	10.1	10.2	10.1		
	PrecAvg	28.5	0.7	0.8	1.9	2.8	4.7	4.1	3.7	3.3	3.1	1.7	1.2	0.7		
Precipitation	PrecMax	48.1	3.4	3.0	8.2	7.4	13.2	10.2	11.5	10.4	13.3	4.8	4.0	3.7		
	PrecMin	17.9	0.0	0.0	0.1	0.0	0.6	0.8	0.2	0.5	0.2	0.0	0.0	0.0		
	PrecStd	6.4	0.6	0.7	1.6	1.6	2.4	2.5	2.4	2.1	2.6	1.2	1.1	0.7		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.4	67.4	77.9	84.5	91.0	97.0	100.0	97.4	94.2	86.3	74.0	59.5		
		MCWB	45.2	48.8	56.4	60.3	67.4	71.7	73.7	74.3	69.4	65.7	54.7	45.9		
	2%	DB	52.8	58.3	71.6	78.6	86.2	92.7	96.4	92.8	89.8	78.8	67.6	53.4		
		MCWB	41.4	45.6	54.2	59.3	66.7	72.2	74.7	73.8	69.5	60.7	51.8	42.9		
	5%	DB	47.2	52.3	65.4	72.8	82.1	89.8	92.8	89.8	85.6	73.6	61.5	48.1		
		MCWB	38.7	42.9	51.8	57.1	65.1	71.7	74.4	73.5	68.4	58.2	49.7	39.9		
	10%	DB	42.5	46.3	59.7	68.0	77.5	85.8	90.0	85.9	81.3	69.6	56.0	43.4		
		MCWB	36.1	39.5	48.7	54.5	62.9	70.1	73.7	72.6	66.1	57.3	46.0	36.8		
	0.4%	WB	46.1	52.4	61.5	64.5	72.2	77.4	79.5	78.9	73.9	68.2	58.7	48.5		
		MCDB	58.7	61.3	69.6	76.7	84.3	88.8	91.5	88.9	86.0	79.2	66.2	57.0		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.1	47.1	58.1	61.3	69.6	75.0	77.7	76.7	72.0	64.7	54.3	43.4		
		MCDB	51.8	57.2	66.7	72.8	81.2	87.0	90.2	87.4	84.0	73.2	63.8	52.0		
	5%	WB	39.2	43.3	53.5	58.7	67.3	73.1	76.3	75.1	70.1	61.1	50.5	40.3		
		MCDB	46.3	51.9	62.8	70.8	78.2	85.3	88.5	85.6	81.7	70.6	60.6	47.7		
	10%	WB	36.4	39.6	49.2	55.9	64.8	71.4	74.8	73.4	68.3	58.3	46.8	37.1		
		MCDB	42.1	46.3	59.8	66.6	74.5	83.5	86.6	83.9	78.5	67.2	56.1	43.1		
Mean Daily Temperature Range	5% DB	MDBR	19.3	19.6	22.8	22.7	20.9	20.8	20.7	20.2	22.6	22.6	21.8	18.5		
		MCDBR	26.9	28.4	32.2	30.3	27.1	25.1	25.3	24.5	26.5	28.7	29.9	26.0		
		MCWBR	17.6	17.8	18.0	15.9	12.3	10.3	10.2	10.4	10.8	14.2	17.2	17.3		
	5% WB	MCDBR	25.2	27.1	28.3	26.7	23.1	22.4	22.0	21.6	22.9	23.0	27.5	24.6		
		MCWBR	17.2	17.6	17.4	15.4	12.4	11.0	10.9	10.8	10.8	13.6	16.9	17.3		
Clear-Sky Solar Irradiance	taub		0.274	0.279	0.306	0.343	0.370	0.383	0.399	0.397	0.361	0.318	0.278	0.269		
	taud		2.489	2.499	2.453	2.368	2.366	2.376	2.352	2.348	2.403	2.508	2.534	2.504		
	Ebn at Noon		286	301	302	295	288	283	277	275	278	282	282	279		
	Edh at Noon		23	27	32	37	38	38	39	38	33	27	23	22		
All-Sky Solar Radiation	RadAvg		707	967	1313	1656	1917	2196	2144	1874	1523	1093	784	601		
	RadStd		65	73	113	93	117	133	104	106	95	107	55	61		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page